

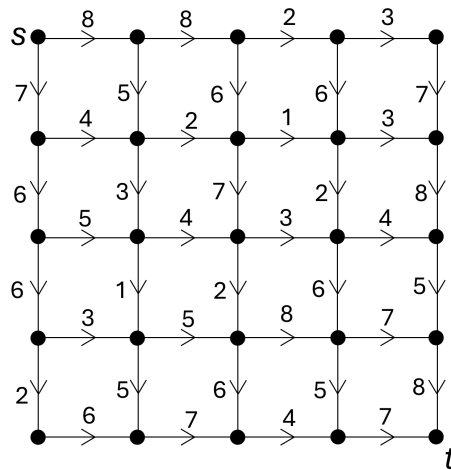
MATH3301 Assignment 2 - Graph Theory

See the Course Profile for the due date. This is not the assignment for MATH7331 students.

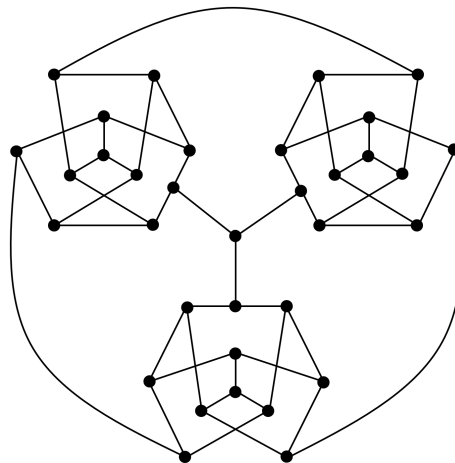
- (1) **(10 marks)** Prove that the Ramsey number $R(3, 3, 4) \leq 34$. You may use any of the following results.

$$R(3, 3) = 6, \quad R(3, 4) = 9, \quad R(4, 4) = 18, \quad R(3, 3, 3) = 17$$

- (2) **(10 marks)** Find a maximum flow and a minimum cut in the network below. The source s , sink t , and capacity of each edge are indicated.



- (3) **(10 marks)** Determine whether the graph shown below is Hamiltonian and justify your answer.



See over page for Question 4.

- (4) **(10 marks)** Complete the following proof that the 5-vertex configuration consisting of a 4-cycle ring and a vertex which is joined to each vertex of the ring is reducible.

Proof: For a contradiction, suppose this configuration is in a minimal counter-example G to the Four Colour Theorem. Let the 4-cycle ring be (r, b, g, y) and let v be the other vertex of the configuration. Since G is a minimal counter-example, $G - \{v\}$ has a proper vertex colouring with 4 colours (joining r to g creates a planar triangulation, and any proper vertex colouring of this triangulation is also a proper vertex colouring of $G - \{v\}$). Moreover, in any such colouring the colours on r, b, g and y are pairwise distinct, because otherwise the colouring can be extended to a proper vertex colouring of G with 4 colours (which is not possible because G is a counter-example). Thus, we can assume without loss of generality that there is a proper vertex colouring of $G - \{v\}$ with colours red, blue, green and yellow such that r is red, b is blue, g is green and y is yellow.

Let $G_{red-green}$ be the subgraph of $G - \{v\}$ induced by the red and green vertices, and let H be the component of $G_{red-green}$ containing r . If H does not contain g , then we can interchange the colours of the vertices of H to obtain a new proper vertex colouring of $G - \{v\}$ which can be completed to a proper vertex colouring of G by colouring v red. This is a contradiction and it follows that H contains g .